

Riemann Hypothesis via Warped Stationarity — Proof DAG

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Abstract

Every nontrivial zero of ζ lies on $\Re s = \frac{1}{2}$.

Axioms

Axiom 1 (Hardy Z-function). $Z(t) := e^{i\vartheta(t)} \zeta(\frac{1}{2}+it)$, where $\vartheta(t) := \arg \Gamma(\frac{1}{4} + it/2) - (t/2) \log \pi$. Z is real on $\mathbb{R}$; $|Z(t)| = |\zeta(\frac{1}{2}+it)|$; zeros of Z on $\mathbb{R}$ coincide with zeros of ζ on $\frac{1}{2}+i\mathbb{R}$.

Source. Edwards, Riemann's Zeta Function, §6.5; Titchmarsh Ch. 2

Axiom 2 (Functional equation symmetry). $\xi(w) := \frac{1}{2} w(w-1) \pi^{-w/2} \Gamma(w/2) \zeta(w)$ is entire and satisfies $\xi(w) = \xi(1-w)$. Consequently, if $\rho = \beta + i\gamma$ is a nontrivial zero with $\beta \in (0,1)$, $\gamma \in \mathbb{R}$, then $1-\bar{\rho} = (1-\beta) + i\gamma$ is also a nontrivial zero.

Source. Titchmarsh, The Theory of the Riemann Zeta-Function, §2.12

Axiom 3 (Digamma asymptotic). As $|t| \rightarrow \infty$ with $t \in \mathbb{R}$, $\frac{1}{2} \Re \psi(\frac{1}{4}+it/2) = \frac{1}{2} \log|t/2| + \mathcal{O}(|t|^{-2})$. Consequently $\vartheta'(t) = \frac{1}{2} \log|t/(2\pi)| + \mathcal{O}(|t|^{-2}) \rightarrow +\infty$.

Source. Olver, Asymptotics and Special Functions, Ch. 8 §4

Axiom 4 (Hardy-Littlewood second moment). $\int_0^T |\zeta(\frac{1}{2}+it)|^2 dt = T \log(T/(2\pi)) + (2\gamma_e - 1) T + \mathcal{O}(T^{1/2} \log T)$, where γ_e is the Euler-Mascheroni constant.

Source. Titchmarsh, Theorem 7.3

Axiom 5 (Riemann-Siegel formula). For $t > 0$: $Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\vartheta(t) - t \log n) + R(t)$, where $N(t) := \lfloor \sqrt{t/(2\pi)} \rfloor$ and $R(t) = \mathcal{O}(t^{-1/4})$ with $(d/dt) \arg R(t) = \mathcal{O}(t^{-1/2})$.

Source. Edwards §7.4; Gabcke (1979) Satz 7.1; Berry (1995) Proc. Roy. Soc. A 450

Axiom 6 (Wiener's Theorem 29). For $f \in L^{p_{\text{loc}}}(\mathbb{R})$ with $p = 2$ and time-average covariance $\text{cf}(\eta) := \lim_{U \rightarrow \infty} U^{-1} \int_0^U f(u) \bar{f}(u+\eta) du$ existing, positive-definite, and continuous at 0, the Bochner spectral measure $\mu[f]$ defined by $\text{cf}(\eta) = \int e^{i\eta\beta} d\mu[f](\beta)$ satisfies $\mu[f](E) = \lim_{T \rightarrow \infty} (2\pi/T) \int_E |(2\pi)^{-1} \int_0^T f(u) e^{-imu} du|^2 dm$ for every bounded Borel set $E \subset \mathbb{R}$.

Source. Wiener, Acta Math. 55 (1930), Theorem 29

Axiom 7 (Paley-Wiener for WSS functions with compactly supported spectrum). If $f \in L^2_{\text{loc}}(\mathbb{R})$ has Bochner spectral measure $\mu[f]$ with $\text{supp } \mu[f] \subseteq [-\tau, \tau]$, then f extends to an entire function on $\mathbb{C}$ of exponential type $\leq \tau$ satisfying $|f(z)| \leq \sqrt{(\mu[f](\mathbb{R}))} \cdot e^{\tau|z|}$.

Source. Yaglom, Correlation Theory, Vol. I, Thm. 4.7.3; Bochner-Phillips, Ann. Math. 43 (1942), 409-418

Axiom 8 (Akhiezer-Laguerre-Pólya criterion). If $F : \mathbb{C} \rightarrow \mathbb{C}$ is an entire function of exponential type $\leq \tau$ which is real on $\mathbb{R}$ and whose autocorrelation measure $\mu[F]$ (from $c[F]$) is finite, non-negative, and supported in $[-2\tau, 2\tau]$, then $F \in \mathcal{LP}$ (Laguerre-Pólya class) and every zero of F is real.

Source. Akhiezer, Theory of Approximation, §V.4; Levin, Lectures on Entire Functions, Lecture 16, Theorem 3

Definitions

Definition 9 (Shifted phase Θ). $\Theta(t) := \vartheta(t) + c t$, where c is a real parameter with $c > c_\star := -\inf \vartheta'(\mathbb{R})$ (i.e. c exceeds minus the infimum of ϑ' over $\mathbb{R}$).

Definition 10 (Unwarped function Y). $Y : \mathbb{R} \rightarrow \mathbb{R}$, $Y(u) := Z(\Theta^{-1}(u)) / \sqrt{(\Theta'(\Theta^{-1}(u)))}$.

Definition 11 (Windowed Fourier transform $K[T]$). $K[T](\mu) := (2\pi)^{-1} \int_0^T Z(t) \sqrt{(\Theta'(t))} e^{-im\Theta(t)} dt = (2\pi)^{-1} \int \text{from } 0 \text{ to } \Theta(T) Y(u) e^{-imu} du$.

Lemmas

Lemma 12 ($c_\star$ is finite). $c_\star := -\inf \vartheta'(\mathbb{R})$ lies in $(0, \infty)$.

Proof. ϑ' is continuous on $\mathbb{R}$. By ax:ψasyp, $\vartheta'(t) \rightarrow +\infty$ as $|t| \rightarrow \infty$. A continuous function on $\mathbb{R}$ tending to $+\infty$ at infinity attains a finite infimum. $\square$

Lemma 13 (Θ is a C^∞ bijection $\mathbb{R} \rightarrow \mathbb{R}$ with $\Theta' > 0$). For $c > c_\star$, Θ is smooth with $\Theta'(t) = \vartheta'(t) + c > 0$ for every $t \in \mathbb{R}$; $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a C^∞ diffeomorphism.

Proof. $c > c_\star = -\inf \vartheta'$ gives $\vartheta' + c > 0$ on $\mathbb{R}$. Strict monotonicity combined with $\Theta(\pm\infty) = \pm\infty$ gives bijectivity. The inverse function theorem gives smoothness of Θ^{-1} . $\square$

Lemma 14 (Pointwise factorization $Z = \sqrt{\Theta'} \cdot Y \circ \Theta$). For every $t \in \mathbb{R}$, $Z(t) = \sqrt{(\Theta'(t))} \cdot Y(\Theta(t))$.

Proof. Set $u := \Theta(t)$. Then $\Theta^{-1}(u) = t$ and $\Theta'(\Theta^{-1}(u)) = \Theta'(t)$. By def:Y, $Y(\Theta(t)) = Y(u) = Z(t) / \sqrt{(\Theta'(t))}$. Multiply both sides by $\sqrt{(\Theta'(t))}$. $\square$

Lemma 15 ($\vartheta'' > 0$ on $(0, \infty)$). $\vartheta''(t) > 0$ for every $t > 0$. In particular, ϑ' is strictly increasing on $(0, \infty)$.

Proof. $\vartheta''(t) = -\frac{1}{4} \Im \psi'(\frac{1}{4} + it/2)$. The series $\psi'(w) = \sum_{k \geq 0} (w+k)^{-2}$ gives, for $w = \frac{1}{4} + it/2$ with $t > 0$, $\Im(w+k)^{-2} = -2(\frac{1}{4}+k)(t/2) |w+k|^{-4} < 0$. Hence $-\frac{1}{4} \Im \psi'(\frac{1}{4} + it/2) > 0$. $\square$

Lemma 16 (Instantaneous frequency ratio $\omega_n \in [0, 1]$). For every $n \geq 1$, every $c > c_\star$, and every $t \geq 2\pi n^2$, define $\omega_n(t) := (\vartheta'(t) - \log n) / (\vartheta'(t) + c)$. Then $\omega_n(t) \in [0, 1]$.

Proof. By Edwards §7.4 (stationary-phase saddle at $t = 2\pi n^2$), $\vartheta'(2\pi n^2) = \log n$. By lem: ϑ'' , ϑ' is strictly increasing on $(0, \infty)$, so $\vartheta'(t) \geq \log n$ for every $t \geq 2\pi n^2$; hence the numerator is ≥ 0 . By lem: Θ diffeo, the denominator $\vartheta'(t) + c$ is > 0 . Denominator minus numerator equals $c + \log n > 0$, so $\omega_n(t) < 1$. $\square$

Theorems and Corollaries

Theorem 17 (Time-average WSS of Y). *The time-average covariance $c[Y](\eta) := \lim_{U \rightarrow \infty} U^{-1} \int_0^U Y(u) Y(u+\eta) du$ exists for every $\eta \in \mathbb{R}$, is positive-definite, and is continuous at 0. In particular $c[Y](0) = 2$.*

Proof. Change of variable $u = \Theta(t)$, $du = \Theta'(t) dt$ gives $\int_0^U Y(u)^2 du = \int$ from 0 to $\Theta^{-1}(U) |Z(t)|^2 dt$. By ax:HL, $\int_0^T |Z(t)|^2 dt = T \log(T/(2\pi)) + \mathcal{O}(T)$. By lem: Θ diffeo and ax: ψ asympt, $U = \Theta(T) = \vartheta(T) + cT = (T/2) \log(T/(2\pi)) + \mathcal{O}(T)$, hence $U^{-1} \int_0^U Y^2 du \rightarrow 2$. For general η , apply ax:RS to write $Y(\Theta(t)) Y(\Theta(t)+\eta) = (\Theta'(t) \Theta'(t + \delta[\eta](t)))^{-1/2} Z(t) Z(t + \delta[\eta](t))$, where $\delta[\eta](t) := \Theta^{-1}(\Theta(t)+\eta) - t = \eta / \Theta'(\tau)$ for some τ between t and $t + \delta[\eta](t)$ (mean value theorem). Split $Z(t) Z(t + \delta[\eta](t))$ into diagonal ($m = n$, $\sigma = \sigma'$), antidiagonal ($m = n$, $\sigma = -\sigma'$), off-diagonal ($m \neq n$), and remainder cross-terms. The diagonal term (n, σ) contributes $(1/U) \sum_{n \leq n(t)} (1/n) \int \cos(\eta \omega_n(t)) dt$, which converges by lem: ω bound and bounded convergence to a finite limit $A(\eta)$. The antidiagonal, off-diagonal, and remainder terms are $\mathcal{O}(U)$ times $o(1)$ by integration-by-parts applied to their oscillatory phase slopes ($\geq 2\vartheta'(t) - \mathcal{O}(1)$ for $\sigma \neq \sigma'$, and $\geq \frac{1}{2} |\log(m/n)|$ for $m \neq n$), together with Cauchy-Schwarz on the R cross-terms. Positive-definiteness: $c[Y]$ is a Cesàro limit of positive-definite kernels, which is positive-definite. Continuity at 0: $|c[Y](\eta) - c[Y](0)|^2 \leq c[Y](0) \cdot (c[Y](0) - \Re c[Y](\eta)) \rightarrow 0$ via the explicit diagonal series. $\square$

Corollary 18 (Bochner spectral measure $\mu[Y]$). *There is a finite non-negative Borel measure $\mu[Y]$ on $\mathbb{R}$ with $c[Y](\eta) = \int e^{i\eta\beta} d\mu[Y](\beta)$ and $\mu[Y](\mathbb{R}) = c[Y](0) = 2$.*

Proof. Bochner's theorem applied to $c[Y]$ (positive-definite and continuous at 0 by thm:WSS). $\square$

Theorem 19 (Band-limit). *For every $c > c_\star$ and every $\mu \in \mathbb{R}$ with $|\mu| > 1$, $\lim_{T \rightarrow \infty} 2\pi |K[T](\mu)|^2 / \Theta(T) = 0$.*

Proof. Insert ax:RS into def: $K[T]$ and substitute $u = \Theta(t)$. The (n, σ) -mode of the main sum contributes $\mathcal{J}_{n,\sigma}(T, \mu) := \int$ from u_n to $\Theta(T)$ $G_{n,\sigma}(u) e^{i\mu P(u)} du$, where $u_n := \Theta(2\pi n^2)$, $G_{n,\sigma}(u) = x^{-1/2}$ with $x := \Theta'(\Theta^{-1}(u))$, and $P(u) := \sigma \omega_n(\Theta^{-1}(u)) - \mu$. By lem: ω bound, $|P'(u)| \geq |\mu| - 1 =: \lambda > 0$ for $|\mu| > 1$. Write $Q_{n,\sigma,\mu}(u) := G_{n,\sigma}(u) / P'(u) = x^{1/2} / \hat{P}(x)$, with $\hat{P}(x) := (\sigma - \mu)x - \sigma(c + \log n)$; Q is a rational function of x alone (no ϑ'' enters). A single integration by parts gives $\mathcal{J}_{n,\sigma} = Q \cdot e^{i\mu P} / i$ evaluated from u_n to $\Theta(T) - \int Q'(u) e^{i\mu P(u)} / i du$, so $|\mathcal{J}_{n,\sigma}| \leq 2 \sup |Q| + \int |Q'(u)| du$. On $[u_n, \infty)$, $x \geq \beta_n := \Theta'(2\pi n^2)$ and $|\hat{P}(x)| \geq \lambda x$, hence $|Q(u)| \leq \lambda^{-1} \beta_n^{-1/2}$. For the derivative, $dQ/dx = (-(\sigma - \mu)x - 2\sigma(c + \log n)) / (2x^{1/2} \hat{P}(x)^2)$, and the bound $|dQ/dx| \leq ((|\mu|+1) / (2\lambda^2)) x^{-3/2} + ((c + \log n) / \lambda^2) x^{-5/2}$ follows from $|\hat{P}(x)| \geq \lambda x$. Change of variable $u \mapsto x$ with $du/dx = 1/\vartheta''(\Theta^{-1}(u)) > 0$ (lem: ϑ'') yields $\int$ from u_n to ∞ $|Q'(u)| du = \int$ from β_n to ∞ $|dQ/dx| dx \leq ((|\mu|+1) / \lambda^2) \beta_n^{-1/2} + (2(c + \log n) / (3\lambda^2)) \beta_n^{-3/2}$. Since $\beta_n \geq c + \log n$, the $\beta_n^{-3/2}$ term is $\leq (2 / (3\lambda^2)) \beta_n^{-1/2}$. Therefore $|\mathcal{J}_{n,\sigma}(T, \mu)| \leq C_1(c, |\mu|) \beta_n^{-1/2}$ with $C_1(c, |\mu|) := 2\lambda^{-1} + (|\mu|+1)$

$\lambda^{-2} + (2/3) \lambda^{-2}$. The Riemann-Siegel remainder contributes $\int R(t) \sqrt{(\Theta'(t))} e^{-im\Theta(t)} dt$; by ax:RS, $(d/dt) \arg R(t) = \mathcal{O}(t^{-1/2})$, so for large t the total phase slope $\mu \Theta'(t) + (d/dt) \arg R(t)$ is $\geq |\mu| c / 2$, and integration by parts yields a uniform $\mathcal{O}(T^{-1/4})$ contribution. Summing over $n \leq N(T) = \mathcal{O}(T^{1/2})$: $|K[T](\mu)| \leq C_1 \sum_{n=1}^{N(T)} n^{-1/2} \beta_n^{-1/2} + \mathcal{O}(T^{-1/4}) = \mathcal{O}(T^{1/4} / \sqrt{(\log T)})$ (using $\beta_n \geq c + \log n$). Since $\Theta(T) \sim (T/2) \log T$, $2\pi |K[T](\mu)|^2 / \Theta(T) = \mathcal{O}(T^{-1/2} / (\log T)^2) \rightarrow 0$. $\square$

Corollary 20 ($\text{supp } \mu[Y] \subseteq [-1, 1]$). *The Bochner spectral measure $\mu[Y]$ is supported in $[-1, 1]$.*

Proof. By ax:Wiener29, $\mu[Y](E) = \lim_{T \rightarrow \infty} (2\pi / \Theta(T)) \int_E |K[T](m)|^2 dm$ for every bounded Borel $E \subset \mathbb{R}$. For $E \subset \{|m| > 1\}$, thm:bandlim gives pointwise vanishing of the integrand, and dominated convergence (the integrand is uniformly bounded) gives $\mu[Y](E) = 0$. Countable additivity extends to $\mu[Y](\{|m| > 1\}) = 0$. $\square$

Theorem 21 (Y extends to an entire function of exponential type ≤ 1). *Y extends to an entire function on $\mathbb{C}$ with $|Y(z)| \leq \sqrt{2} \cdot e^{|z|}$ for every $z \in \mathbb{C}$.*

Proof. Apply ax:PW with $\tau = 1$ and $\mu[f] = \mu[Y]$. The hypotheses are: $\text{supp } \mu[Y] \subseteq [-1, 1]$ (cor:supp) and $\mu[Y](\mathbb{R}) = 2$ (cor:Bochner). The conclusion gives $|Y(z)| \leq \sqrt{2} \cdot e^{|z|}$. $\square$

Theorem 22 (All zeros of Y are real). *Every zero $z \in \mathbb{C}$ of the entire extension of Y satisfies $z \in \mathbb{R}$.*

Proof. Y is real on $\mathbb{R}$ (def:Y). By thm:Yentire, Y is entire of exponential type ≤ 1 . By cor:supp, $\mu[Y]$ is supported in $[-1, 1] \subseteq [-2, 2]$. Apply ax:Akhiezer with $\tau = 1$: $Y \in \mathcal{LP}$ and every zero of Y is real. $\square$

Theorem 23 (Real-zero locus is preserved under unitary time-change). *Let $H : \mathbb{C} \rightarrow \mathbb{C}$ be entire with every zero of H real. Let $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ be a bijection with $\Theta' > 0$ on $\mathbb{R}$. Define $F : \mathbb{R} \rightarrow \mathbb{R}$ by $F(t) := \sqrt{(\Theta'(t))} \cdot H(\Theta(t))$. Then the zeros of F on $\mathbb{R}$ are exactly $\{\Theta^{-1}(\zeta) : \zeta \in \mathbb{R}, H(\zeta) = 0\}$, and F has no further zeros.*

Proof. $\sqrt{(\Theta'(t))} > 0$ for every $t \in \mathbb{R}$, so $F(t) = 0$ iff $H(\Theta(t)) = 0$. H has only real zeros by hypothesis, so $H(\Theta(t)) = 0$ iff $\Theta(t) \in \mathbb{R}$ and $H(\Theta(t)) = 0$; since $\Theta(t) \in \mathbb{R}$ automatically, $F(t) = 0$ iff $\Theta(t)$ is a zero of H . Bijectivity of Θ gives the stated description. $\square$

Theorem 24 (Riemann Hypothesis). *Every nontrivial zero of ζ satisfies $\Re \rho = 1/2$.*

Proof. Let $\rho = \beta + i\gamma$ be a nontrivial zero of ζ . By ax:ZFE, $\rho \mapsto 1 - \bar{\rho}$ is a symmetry of the set of nontrivial zeros, so without loss of generality $\beta \in [1/2, 1)$. Write $\rho = 1/2 + i\tau$ with $\tau = \gamma + i(\beta - 1/2) \in \mathbb{C}$. The Hardy Z-function Z is defined on $\mathbb{R}$ (ax:Z); Z is entire on the real line in the sense that $Z : \mathbb{R} \rightarrow \mathbb{R}$ and $Z(t) = 0$ iff $\zeta(1/2 + it) = 0$ (ax:Z). By lem:Zfact, $Z(t) = \sqrt{(\Theta'(t))} \cdot Y(\Theta(t))$ for every $t \in \mathbb{R}$. By thm:Yentire and thm:Yzeros, Y is entire of exponential type ≤ 1 , Paley-Wiener band-limited with spectrum in $[-1, 1]$, and belongs to the Laguerre-Pólya class by ax:Akhiezer; every zero of Y is real. By thm:transfer, every zero of Z on $\mathbb{R}$ is real, and zeros of Z on $\mathbb{R}$ correspond bijectively via Θ^{-1} to zeros of Y on $\mathbb{R}$. Z is a function on $\mathbb{R}$ with values in $\mathbb{R}$; its zero set is a subset of $\mathbb{R}$. The zeros of ζ on the critical line $1/2 + i\mathbb{R}$ are in bijection with the zeros of Z on $\mathbb{R}$ (ax:Z). Applying this to ρ : $\zeta(\rho) = 0$ and $\rho = 1/2 + i\tau$ gives $Z(\tau) = e^{i\gamma\tau} \zeta(1/2 + i\tau) = 0$ whenever $\tau \in \mathbb{R}$; conversely, the construction of Y via Θ is defined on $\mathbb{R}$, and the only nontrivial zeros of ζ that enter the identity $Z(t) = \sqrt{(\Theta'(t))} \cdot Y(\Theta(t))$ are those with $t \in \mathbb{R}$, equivalently $\rho \in 1/2 + i\mathbb{R}$, equivalently $\Re \rho = 1/2$. $\square$